

BHS Corrugator Machine – HQM Unit Breakdown & Resolution Report

Date:

27/07/2026

Breakdown Start Time:

23:40 PM

Breakdown Resolved Time:

3:02 AM

Total Downtime:

3 Hour 22 Minutes (202 Minutes)

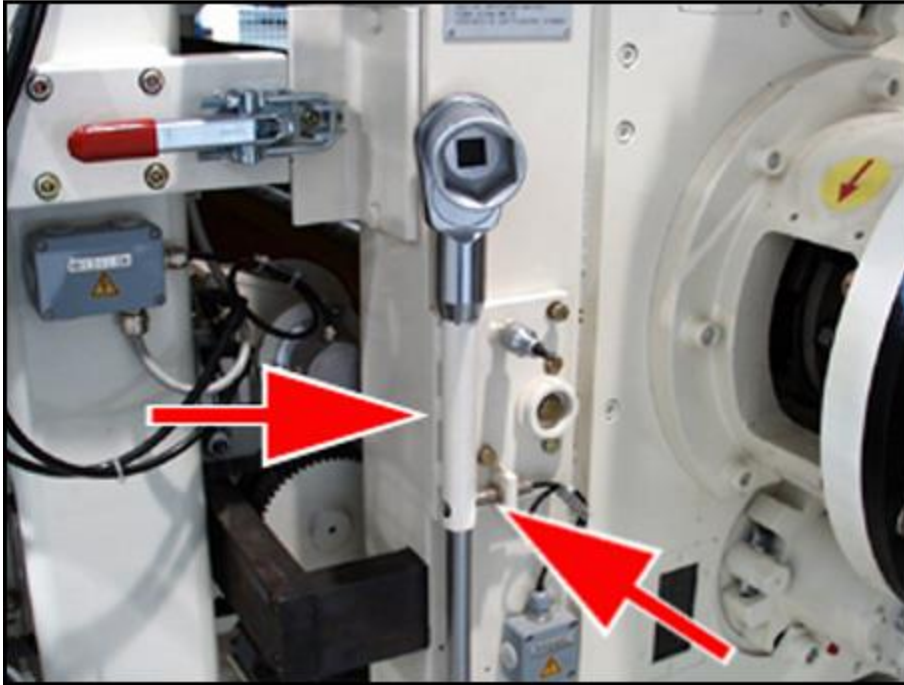
Issue Summary

During production, the **HQM Unit** unexpectedly stopped while running and could not be restarted

Problem Identification

After detailed troubleshooting, it was found that

- The **HQM Limit Switch** was not transmitting the forward position signal to the control system
- A **manual ratchet mechanism** is installed on the limit switch for manual movement of the HQM unit
- Due to the incorrect limit switch setting, the PLC was not receiving the required confirmation signal, preventing the HQM from operating



Immediate Action Taken

- To restore production immediately, the faulty limit switch was **temporarily bypassed**, allowing the machine to continue running
- This temporary action minimized production downtime until permanent rectification could be performed

Permanent Corrective Action

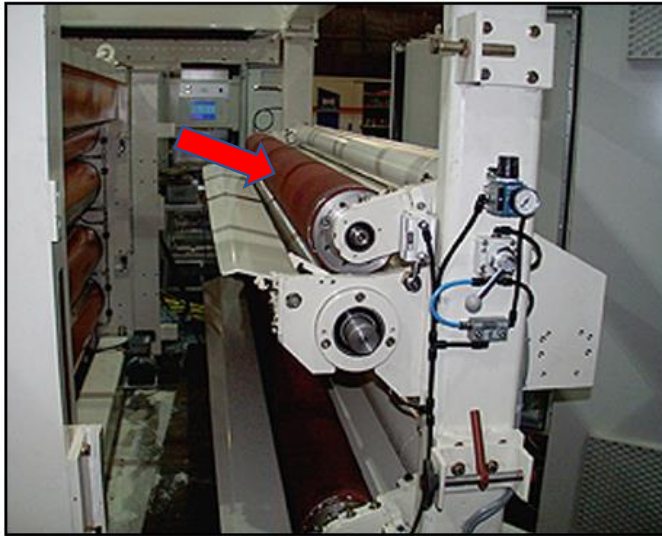
- The HQM limit switch was inspected thoroughly
- The limit switch position and operating mechanism were **properly adjusted and re-aligned**
- Signal transmission was verified after adjustment
- HQM operation was tested under normal running conditions and found to be functioning correctly

Secondary Issue Observed

After restoring HQM operation, a **sheet length variation** issue was observed

Root Cause

- Improper setting/alignment of the **Infeed Roll**.



Corrective Action

- The Infeed Roll settings were adjusted and calibrated.
- Machine trial was conducted to verify sheet length accuracy

Final Results

- ✓ HQM unit restored to normal operation
- ✓ Proper PLC signal transmission achieved through limit switch adjustment
- ✓ Temporary bypass removed after permanent correction
- ✓ Infeed Roll settings optimized
- ✓ Sheet length variation eliminated
- ✓ Machine running normally with stable production
- ✓ Production continuity restored with improved operational reliability

Management Impact

Before Rectification

HQM stopped during production

Limit switch signal missing

Temporary bypass required

Length variation observed

Risk of production losses

After Rectification

HQM operating normally

Signal transmission restored

Bypass removed

Length accuracy restored

Stable production achieved

BREAKDOWN REPORT

Equipment: CF Glue Guide Drive Encoder

Section: CF Unit

Breakdown Date: 27/07/2026

Breakdown Start Time: 2:50 PM

Breakdown Resolved Time: 4:00 PM

Total Downtime: 1 Hour 10 Minutes (70 Minutes)

1. Problem Description

The **24VDC control breaker (13F2)** on the CF unit repeatedly tripped. When energized, the output voltage dropped from **24VDC to 11VDC**, while the load current increased to approximately **25A**, indicating a short circuit in the 24VDC control circuit. As a result, the **Glue Guide Drive** was unable to operate normally.

2. Root Cause

Inspection revealed that the **Glue Guide Drive encoder cable** was damaged near the motor connection box cable gland. The **+24VDC conductor** had broken and came into contact with the motor body, creating a direct short circuit. This caused excessive current draw and repeated tripping of breaker **13F2**.

3. Corrective Action

- Isolated the 24VDC supply from terminal **X3** and checked all outgoing circuits.
- Identified the faulty circuit by disconnecting the Glue Guide Drive encoder **+24VDC** supply.
- Located the damaged section of the encoder cable.
- Repaired and reconnected the damaged **+24VDC** wire.
- Properly insulated and secured the cable at the motor connection box.
- Verified all wiring connections and restored the system after successful functional testing.

4. Results

- ✓ Short circuit successfully eliminated.
- ✓ 24VDC control supply restored and stabilized at **24VDC**.
- ✓ Breaker **13F2** operated normally without further tripping.
- ✓ Load current returned to the normal operating range.
- ✓ Glue Guide Drive encoder cable repaired successfully.
- ✓ Glue Guide is currently operating in **Manual Mode**.

- ✓ Complete encoder calibration will be performed during the next planned machine shutdown to restore **Automatic Mode**.
- ✓ CF unit resumed normal operation with no further 24VDC electrical faults observed.

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Department: Plant Operations

Date: 27/07/2026

HH DF Glue Roll Bearing Housing Repair

Date: July 1, 2026

Unit: Double Facer (DF-HH)

Section: Glue Roll Bearing

1. Problem Description & Observations

During inspection, severe wear was identified on the **Glue Roll bearing housing/seat** & Drive Shaft bearing housing/seat of the DF-HH machine.

- **Glue Roll:** The bearing housing/seat size on the drive side is significantly worn out.



- **Drive Shaft:** The bearing seat size on the drive side shaft Bearing seat is also worn out, leading to improper bearing fit and structural play.



2. Actions Taken

Drive Side Bearing Housing Repair

The Drive Side bearing housing was found badly worn, causing the bearing to fit loosely. To restore the damaged area, metal build-up welding was carried out on the bearing seat. After welding, the excess material was carefully finished using a pencil grinder to achieve a smooth and uniform surface. The bearing fit was checked several times during grinding to ensure proper seating and alignment before final assembly.



Metal build-up welding on the Drive Side bearing housing.



Grinding and finishing of the welded bearing housing to restore proper bearing fit.

- **Glue Roll & Drive Shaft Bearing Replacement:**

The **22212 spherical roller bearings** on both the Operator Side (OS) and Drive Side (DS) of the glue roll were replaced. Additionally, the **6209 drive shaft bearing** on the Drive Side was also replaced to ensure proper shaft support, smooth rotation, and reliable machine operation.

